

MAHANAND ADIMULAM

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EDUCATION

Rochester Institute of Technology	Dec 2024
Master of Science in Data Science	GPA-3.87/4
Kalinga Institute of Industrial Technology,	May 2022
Bachelor of Technology in Electronics & Computer Science Engineering (Honors)	GPA-3.6/4

EXPERIENCE

Rochester Institute of Technology	Rochester, NY
Python and Java Tutor	Jan 2023 – Dec 2024
- Enhanced comprehension of complex programming concepts in Python and Java for over 90 students by adapting teaching methods to meet diverse learning styles. - Leveraged strong communication skills to clarify complex programming concepts in Python and Java, fostering independent learning and significantly enhancing students' coding capabilities	
HighRadius	India
Full Stack Data Scientist Intern	Jan 2021 – Mar 2021
- Developed an Automated Invoice Management System to refine B2B operations servicing over 5000 invoices by creating a full-stack Invoice Management Application with HTML, CSS, React, Java, and SQL. - Increased data accessibility and analysis speed by managing over 5000 records in a MySQL database on a MySQL server, utilizing SQL for querying and real-time integration through JDBC. - Achieved an 89% forecasting accuracy for invoice close dates through the implementation of a Random Forest machine learning algorithm using Python, enabling instant predictions for both existing and new invoices. - Achieved enhanced decision-making capabilities through the development of a real-time dashboard in Tableau, displaying KPIs like total outstanding receivables, DSO, Aging Analysis, and Payment Trends.	

PROJECTS

Deepfake Slayer (GitHub) [PyTorch, DataLoader, Torchvision, Transforms, Torch, Matplotlib]
- Reimplemented the FFD model in PyTorch, achieving 96% accuracy in classifying defake and real images and generate masks highlighting altered regions in defake images with 98% accuracy. - Engineered a robust data loader using PyTorch to seamlessly handle the intricate FaceForensics++ dataset, ensuring data integrity and accuracy while processing over 2 TB of video data from a research computing database. - Developed an uncertainty estimator over the FFD model, quantifying the confidence level in the model's predictions, resulting in improved trustworthiness of the classification results.
Multi-Headed Model for Toxic Comment Classification (GitHub) [Python, TensorFlow, TextVectorization, LSTM, Bidirectional]
- Leveraged TensorFlow and LSTM Bi-directional layers to effectively classify online comments into threats, obscenities, insults, and identity-based hate speech, achieving a precision of 0.91 and recall of 0.92. - Enhanced model performance through meticulous preprocessing and feature engineering, resulting in a robust training set for accurate classification. - Improved user interaction with the toxicity classification model by designing a user-friendly application utilizing Gradio, facilitating immediate feedback for users on comment toxicity across various categories.
Credit Card Fraud Detection using Autoencoders (GitHub) [Python, Tensorflow, Keras, Matplotlib, Numpy, Pandas, Autoencoder]
- Achieved an AUC score of 0.95 in credit card fraud detection, indicating the model's high performance in distinguishing between fraudulent and legitimate transactions by utilizing Autoencoders with Python and TensorFlow. - Demonstrated expertise in handling a highly imbalanced dataset, where fraudulent transactions represented only 0.172% of the total, resulting in effective detection despite class imbalance. - Optimized the fraud detection model, achieving a 97.77% true positive rate by identifying 55,594 fraudulent transactions out of 56,864, while ensuring a precision rate of 99.97% and limiting false positives to 17. Reduced unnecessary alerts and potential customer inconvenience by achieving a false positive rate of 17.35% through a balanced approach. - Maintained an overall accuracy of 97.74% across a dataset of 56,962 transactions by carefully balancing the detection of fraudulent transactions against the minimization of false alerts.

SKILLS

Programming Languages: Python, Java, R, C++
Databases: SQL, MongoDB, PostgreSQL, Neo4j, SQLite, Apache Spark
Machine Learning (ML): Scikit-learn, TensorFlow, PyTorch, Keras, Pandas, Numpy, OpenCV, BeautifulSoup, NLTK, Gradio
Data Analysis and Statistical Analysis: JMP, Minitab, Tableau, QGIS, Seaborn, Matplotlib, MS Excel, Power BI
Tools and IDE: Jupyter Notebook, PyCharm, Google colab, Git, SQL Workbench, Jira